Smart Travel Companion

SWE-A Project

B.Sc. Computer Science

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  The used AI-based writing tool is ChatGPT (OpenAI), which is available at https://chat.openai.

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1 Introduction

1.1 Idea

Our Main idea was a Smart-Glasses-Application, which allows tourists to experience a well guided and interesting holiday. To ensure this, we wanted to develop an App which guides you through the city of your stay and shows you all the interesting things it has to offer.

1.2 Product

1.2.1 AR Building Recognition and Historical Information

Going sight-seeing seems to have become the new trend when going into foreign waters, especially when the target location has to offer beautiful and interesting structures.

Most of these stunning structures have one thing in common: Their history. Because of this, most cities have established a tour-guide culture, where you listen to the history of all those beautiful structures by listening to a tour guide who is barely understandable while talking about things you don’t want to know. To solve this age old problem, we have come up with a smart and modern solution for such an old problem.

Our Application connects to your Smart Glasses and enables them to render everything you want to know about those interisting structures and monuments right infront of your eyes; understandable and clear.

Once set up, our system automatically registers when you are looking at a known building, instantly displaying you everything you want and need to know about what you are looking at.

1.2.2 Guided AR Route Navigation

To fully experience all the beautiful monuments and points of interest your destination has to offer, you first need to get there - and not just quickly, but safely and enjoyably. The route you take should provide a pleasant journey as well as ensure your safety, which is especially important in countries where unfamiliar surroundings can pose challenges for tourists.

To address this, we developed a Guided AR Route Navigation System that identifies the route best suited for your stay. The system combines multiple factors such as safety, beauty, and accessibility to generate the optimal path. Rather than simply providing the shortest or fastest route, it prioritizes paths that offer cultural highlights, pedestrian-friendly streets, and visually appealing surroundings, enhancing the overall travel experience.

Additionally, the system adapts in real-time, considering dynamic conditions such as traffic, temporary road closures, or areas with higher security risks. By integrating AR overlays, it also guides users visually, highlighting points of interest, providing contextual information, and suggesting route changes that enrich the journey. [1]

Finally, this approach transforms navigation from a mere utility into an exploration experience, ensuring that every step of the journey contributes to both safety and enjoyment, allowing tourists to immerse themselves fully in the destination.

1.2.3 App Loggin with Iris

Having an Account to store the data and preferences of the User is a standard of today's internet society. It's crucial for satisfying a fast and simple experience when transitioning from one device to another, or when restoring data that was lost.

[1] The second and third paragraph were written by the authors and then enhanced by ChatGPT (OpenAI)

To implement this, the user must log in. This is often satisfied with an E-Mail Address and / or username in combination with a password. This, however, can take its time and be frustrating if this data is not at hand, and the time does not allow for calling your mother back home, to ask her to look up your password in some hidden folder.

To solve this modern problem, a modern solution is needed. In this case, the modern solution is an Iris Scanner. To shorten up the time it takes to Log-In, and to keep people from stealing your password, A mandatory or optional Iris Scan can be selected as a Log-In option. This will scan your Iris and save it at our secure databases, so that in the future, you can put on your Smart Glasses and experience our Smart Features seamlessly, and without unnecessary Log-In procedures.

Finally, because we are very interested in protecting the user's data, we ensure that data collected by the Iris Scan Feature will not be sold or used in any way that the user did not consent to.

2 Contribution

Use Case ID	Name	Members
UC - 1	AR Building Recognition and Historical Information	Delil Celic 1552139 Faycal El Amin 1551439
UC - 2	Guided AR Route Navigation	Luca De Simone 1592157 Elissar Abdullah 1501054
UC - 3	App Login with Iris	Luca De Simone 1592157

3 Implementation

3.1 Use Case Diagram

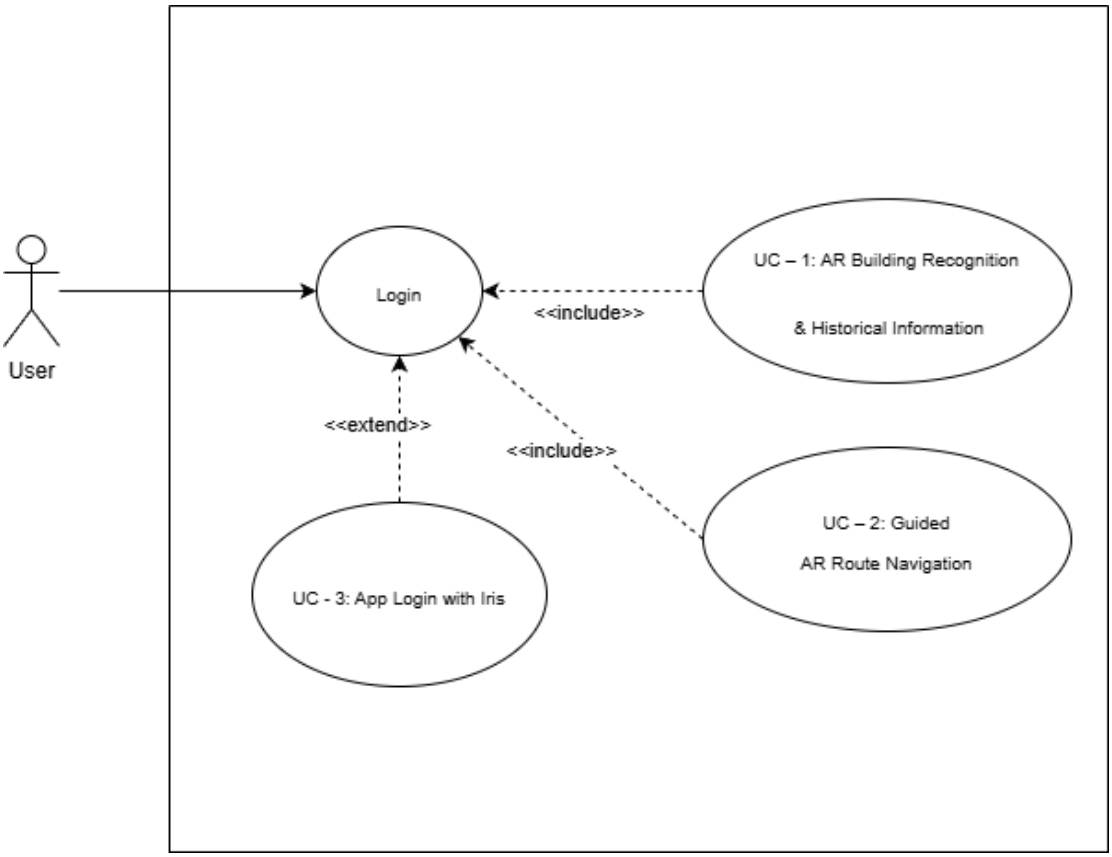

Figure 1: Use-Case Diagram

3.2 Class Diagram

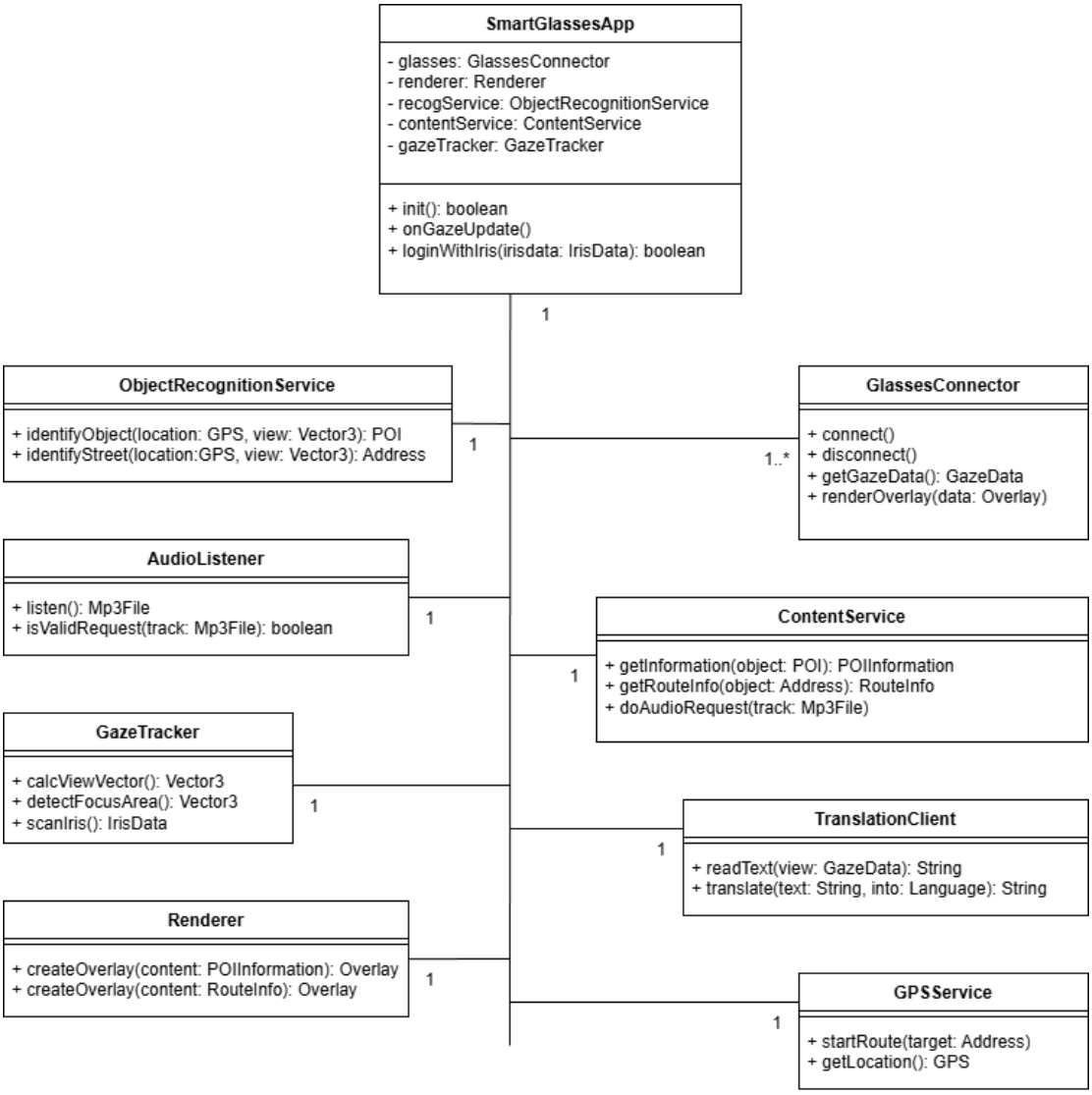

Figure 2: Class Diagram

3.3 UC – 1: AR Building Recognition & Historical Information

3.3.1 Use Case Specification

Name			AR Building Recognition and Historical Information
ID			U1
Description			The system automatically recognizes the building the user is looking at and displays the buildings name and historical information as an augment reality overlay on the smart glasses
Trigger			Stable gaze detected and the user is looking at a building
Actors			User
Pre-conditions			logged in and GPS is functional
Post-conditions			Historical Information about the building is displayed on the smart glasses
Basic Flow			
	Description		main flow of events for successfully recognizing a building and displaying historical information in AR
	Actions		
		1	User looks at a building
		2	The system detects a stable gaze by using eye tracking
		3	The system collects camera data, current GPS location and the viewing direction
		4	The system identifies the building based on collected data
		5	The system requests historical information related to the identified building
		6	The system retrieves the historical data
		7	The system displays the buildings name and historical information as an AR overlay on the smart glasses
Alternative Flow			
	Description		Historical information not available
	Actions		
		1	The system requests historical information related to the building
		2	No historical data is found
		3	An error message indicating that historical information cannot be retrieved is displayed on the smart glasses
Alternative Flow			
	Description		GPS is not available
	Actions		
		1	The system not to retrieve the current GPS location
		2	The system does not receive a valid location signal
		3	The system determines that building recognition cannot be performed without location data
		4	The system displays a notification on the smart glasses indication that GPS is not available
Alternative Flow			
	Description		Building not recognized
	Actions		
		1	The system attempts to identify the building based on the eye tracking
		2	The building cannot be identifed with certainty
		4	The system displays a notification on the smart glasses that the building cannot be identified

Figure 3: UC-1 Specification Table

3.3.2 Activity Diagram

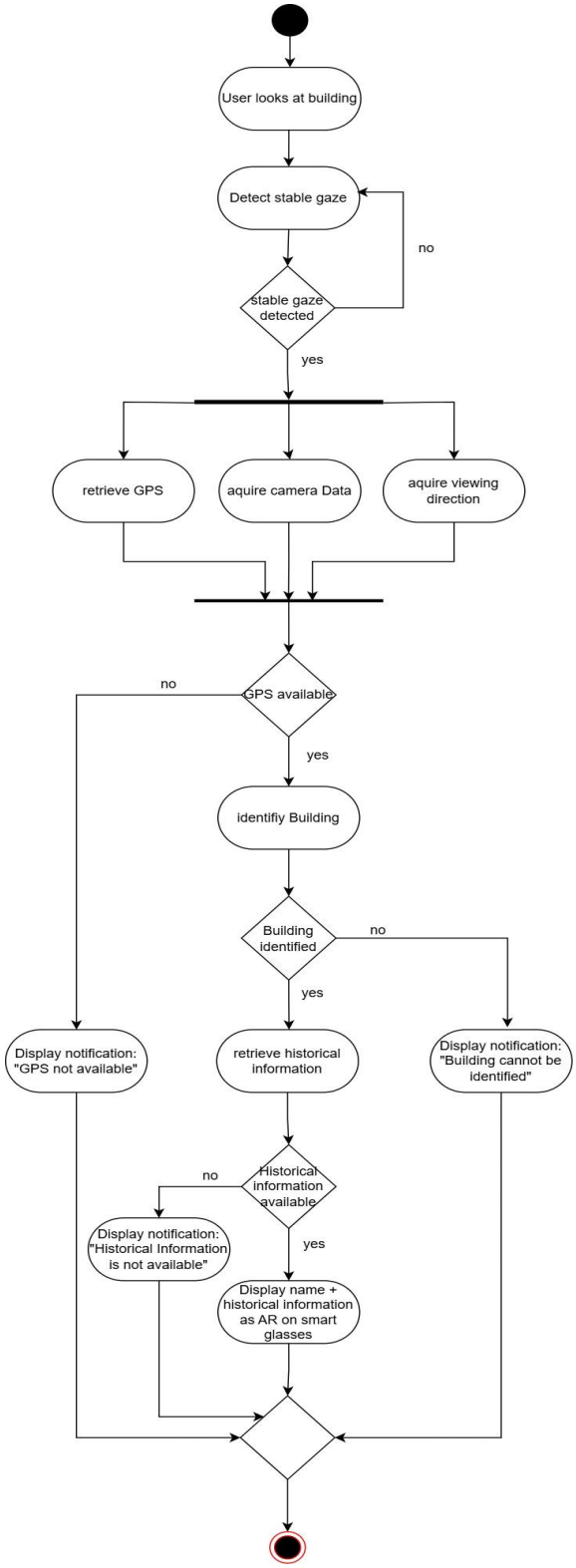

Figure 4: UC-1 Activity Diagram

3.3.3 Sequence Diagram

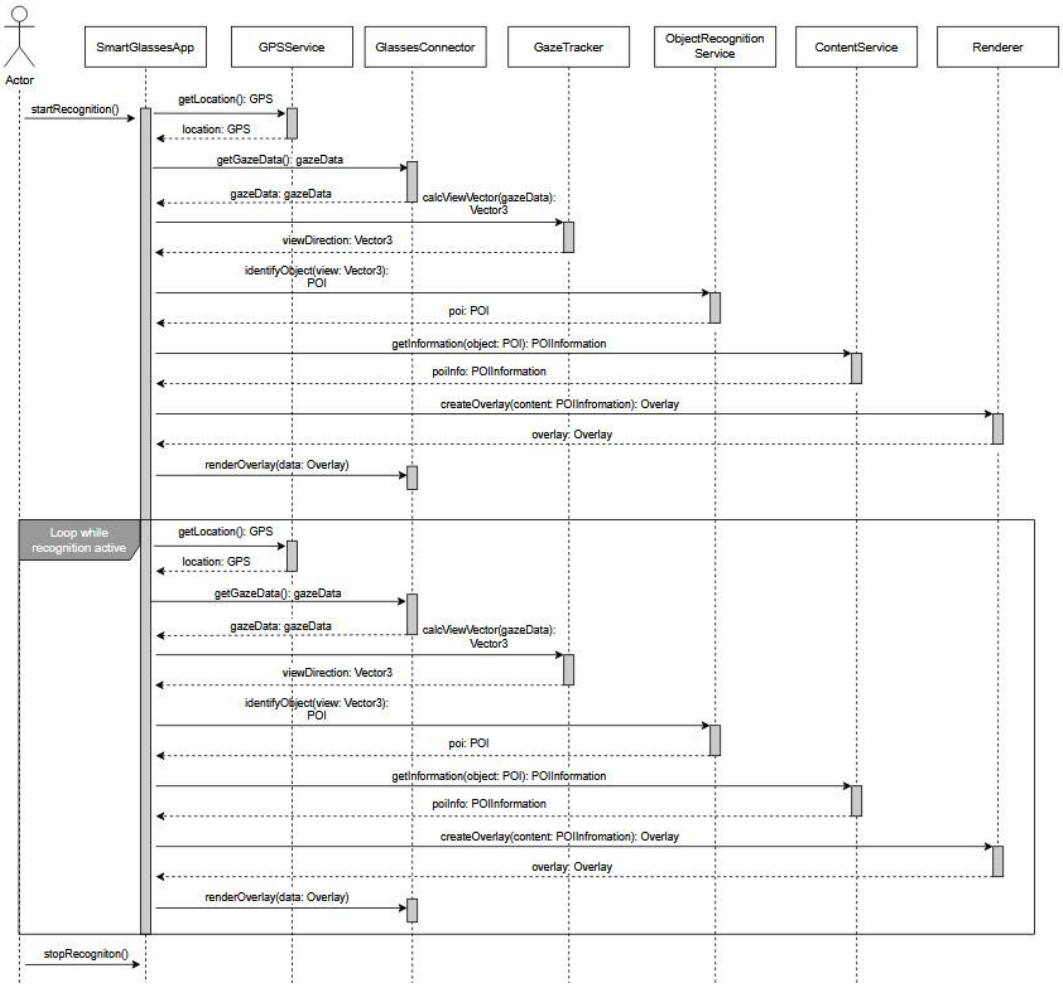

Figure 5: UC-1 Sequence Diagram

3.3.4 GUI Mock-Up

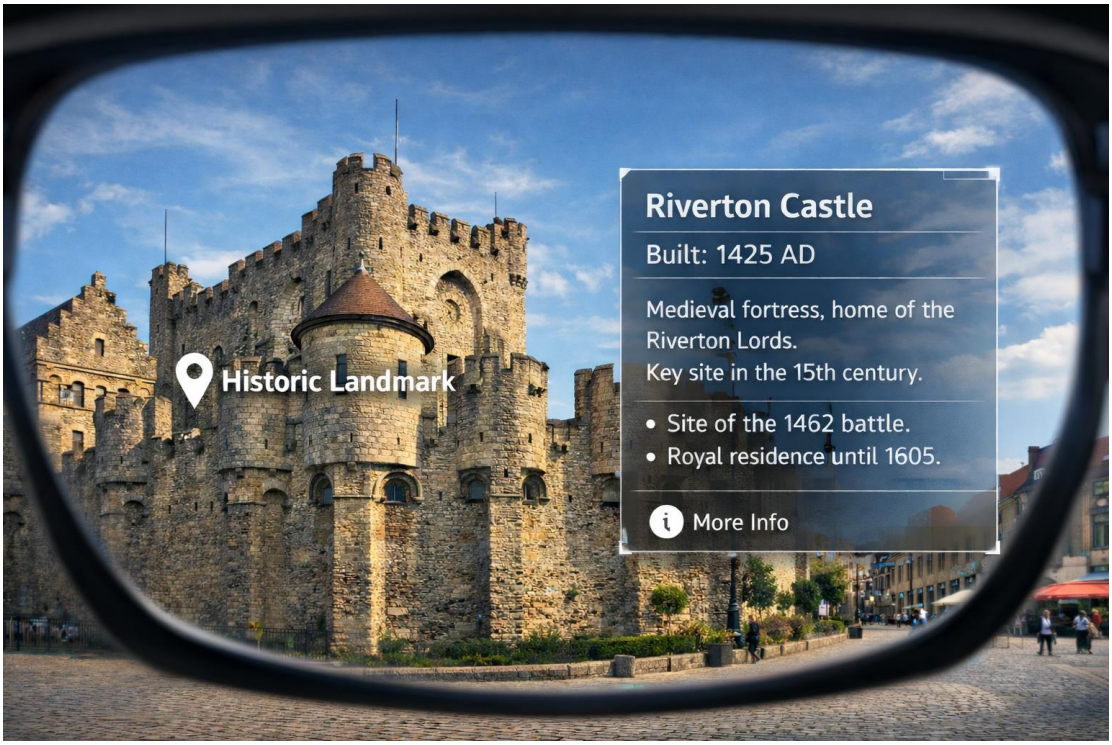

Figure 6: UC-1 GUI Mock-Up[2]

[2] This Image was generated using Artificial Intelligence. The prompt can be found in the AI Prompt Table.

3.3.5 Technical Description

This UC (Use case) allows the user to retrieve information about buildings they are looking at, displayed on their Smart Glasses.

To implement this, the geographical location as well as an image of their FOV (Field of view) are retrieved. We call this their **Gaze Data**. This data is then sent to our servers via the user's app, which then evaluate what the user is looking at based on our algorithms.

Once the POI (Point of Interest) the user is looking at has been established, information, such as historical data and reviews, are packaged and send off back to the user. Upon retrieving the data, the App renders the information on the Smart Glasses.

This process is continuously repeated, automatically rendering the information without requiring action from the user, ignoring cases where no data could be retrieved.

3.4 UC – 2: Guided AR Route Navigation

3.4.1 Use Case Specification

Name	Guided AR Route Navigation
ID	UC-2
Description	The system calculates and renders a guided navigation route on the smart glasses, allowing the user to navigate to a selected destination using augmented reality. The route is optimized for pedestrians and is suitable for tourism and general navigation purposes.
Trigger	The user selects a destination or point of interest and starts navigation in the App
Actors	User
Pre-conditions	User is logged in
Post-conditions	The Navigation is completed or canceled.
Basic Flow	
Description	This is the main scenario, where the user requests navigation and follows route displayed on the smart glasses.
Actions	
1	The user selects a destination via the App.
2	The system calculates an optimal pedestrian route to the destination, based on the Users
3	The system renders the route as augmented reality elements (arrows, path, distance markers) in the user's field of view.
4	The user selects a destination
5	The user follows the visual navigation instructions.
6	The system continuously updates the route based on the user's movement.
7	The system notifies the user when the destination is reached.
Alternative Flow	A
Description	The Destination is unreachable
Actions	
1	The user selects a destination via the App.
2	The system informs the user that navigation is not possible.
3	The Navigation is canceled
Alternative Flow	B
Description	The User Deviates from the Route
Actions	
1	The system detects that the user is off the planned route.
2	The system recalculates the route from the current location.
3	The updated route is rendered on the smart glasses.
4	The user continues navigation.
Alternative Flow	C
Description	The User cancels the Navigation
Actions	
1	The user cancels navigation.
2	The system removes the route from the display.

Figure 7: UC-2 Specification Table

3.4.2 Activity Diagram

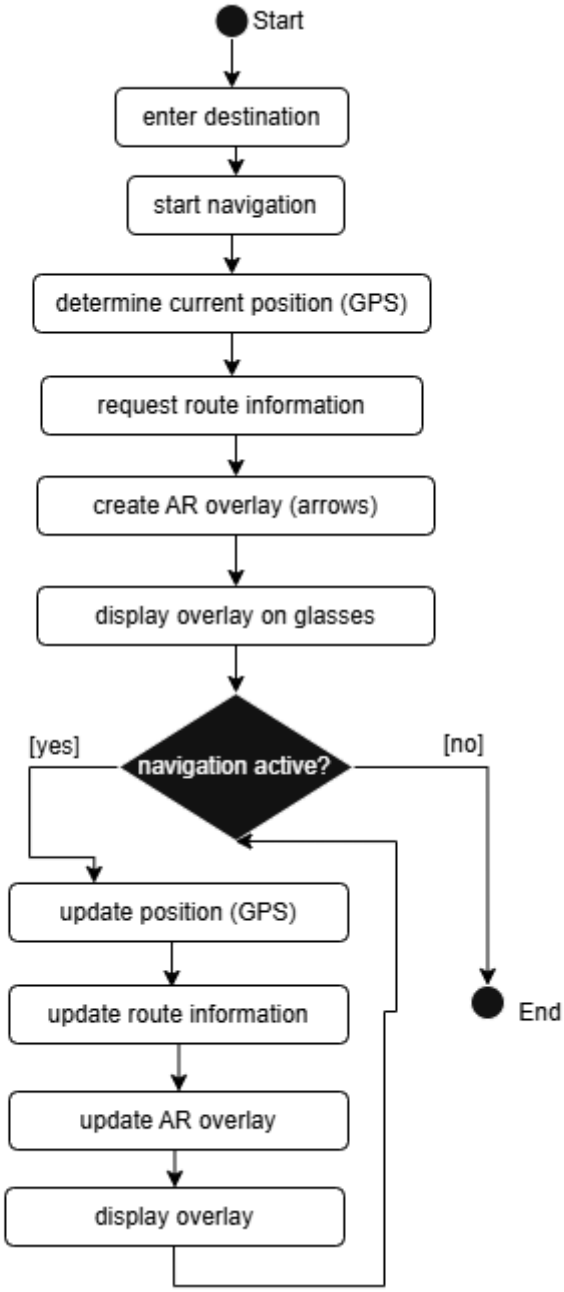

Figure 8: UC-2 Activity Diagram

3.4.3 Sequence Diagram

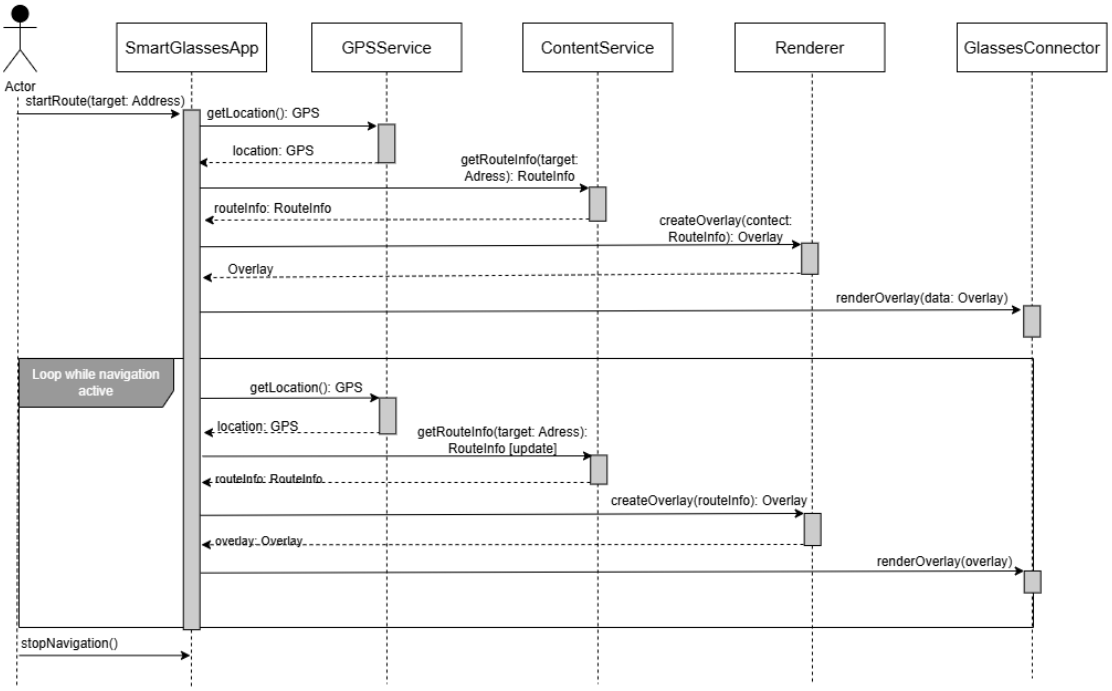

Figure 9: UC-2 Sequence Diagram

3.4.4 GUI Mock-up

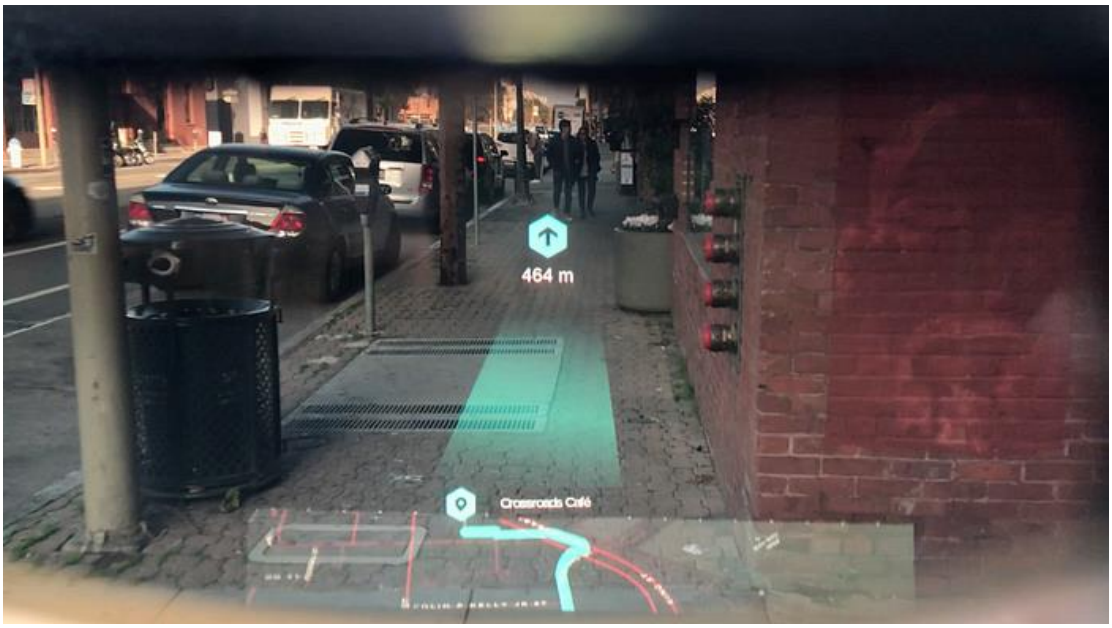

Figure 10: UC-2 GUI Mock-Up

Source: https://miro.medium.com/v2/resize:fit:720/format:webp/0*MqJFT-uDEa1-Crmv.png

3.4.5 Technical Description

This UC navigates the user to their desired destination by rendering the route right in front of them.

This is done by first establishing the user's location and their desired destination via the App. Once this is done, the data is sent to our servers, where our algorithms calculate the safest and most pleasing route for the user, respecting environment and possible danger.

Once this is done, the route data is repeatedly sent to the user, updating in real time based on the new GPS location of the user. The App then renders the path right on the Smart Glasses, augmenting reality by placing it right on the ground.

Should this route not be up to the users' preferences, the ability to cancel or modify the route is available in the App.

3.5 UC – 3: App Loggin with Iris

3.5.1 Use Case Specification

Name	App Login with Iris
ID	UC-3
Description	The system uses an Iris Scanner to scan the eye of the user wearing the Smart Glasses, in order to confirm their identity.
Trigger	The user puts the Smart Glasses on.
Actors	user
Pre-conditions	The used Smart Glasses have a functioning Iris Scanner and are turned on.
Post-conditions	The user is logged in / The Log-In fails (Success / Failure)
Basic Flow	
Description	This is the main scenario, where the user successfully confirms their identity via the Iris Scanner.
Actions	
1	The user puts the Smart Glasses on.
2	The system uses the Iris Scanner to scan the users eye
3	The identity of the user is confirmed successfully.
4	The system logs the user in.
Alternative Flow	A
Description	Identity Verification Failed
Actions	
1	The user puts the Smart Glasses on.
2	The system uses the Iris Scanner to scan the users eye
3	The scan fails / cannot regognize the expected user.
4	The Log-In fails.

Figure 11: UC-3 Specification Table

3.5.2 Activity Diagram

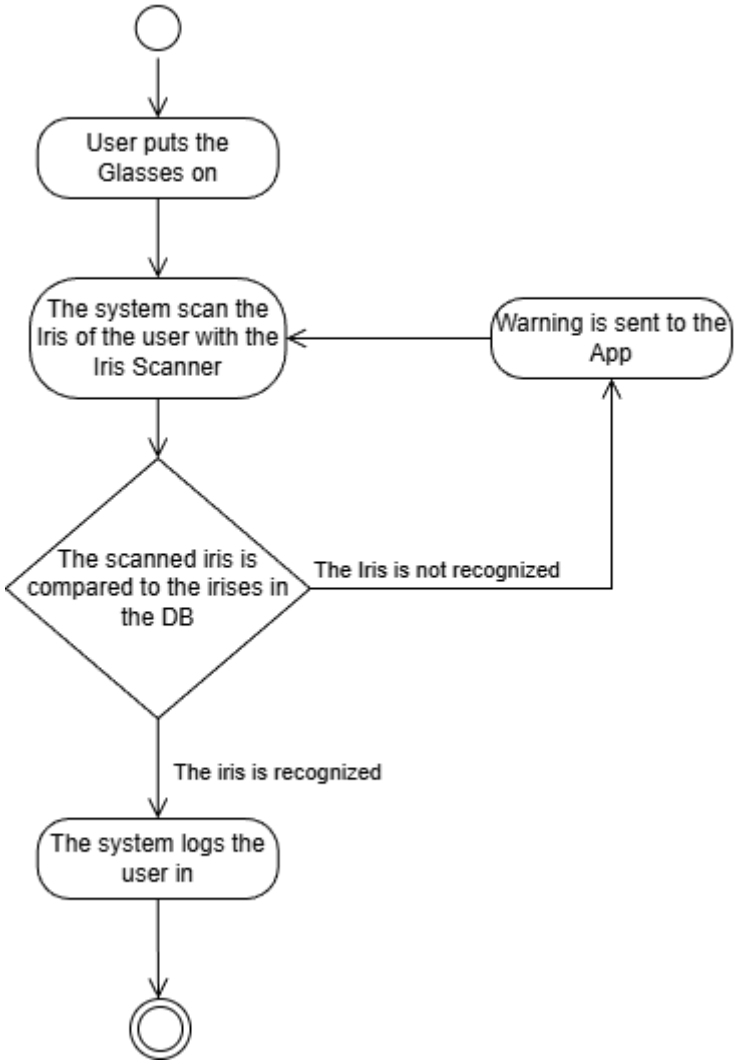

Figure 12: UC-3 Activity Diagram

3.5.3 Sequence Diagram

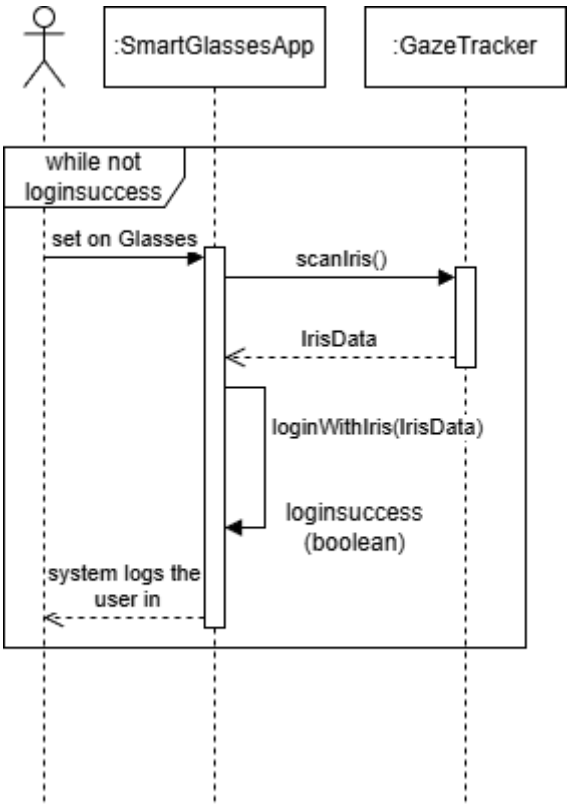

Figure 13: UC-3 Sequence Diagram

3.5.4 Technical Description

This UC allows the user to log into the App by just putting on the Smart Glasses. Upon activating this option in the App, the system tries to scan the user's iris upon startup, waiting until the identity of the user is confirmed, or until the user manually logs into the App via its menu.

To set this up, the user is instructed to wear the Smart Glasses, allowing the system to scan their iris and store it on our databases as reference for later log-ins. Multiple users may be registered in the App, letting them share the device, while maintaining the same account.

This feature also allows parents to monitor the usage of the Smart Glasses App for each individual child.

4 SCRUM Meetings

Monday, 15. Dezember 2025

Meeting Protocol #1

Sprint Planning Meeting

Attendants / Roles

Faycal El Amin – Product owner

Delil Celik – Scrum Master

Elissar Abdullah – Developer

Luca De Simone – Developer

Discussion Points and Decision:

1) Sprint Goal: The goal of this sprint was to define the core ideas of the project and agree on the overall vision of the Smart Travel Companion.
    - The team discussed the general problem space of tourism and navigation in unfamiliar cities
    - It was agreed to develop a Smart Glasses application that supports tourists using augmented reality

2) Sprint Backlog:
    - Define project idea and vision
    - Identify the target users(tourists)
    - Define the main value of the application
    -
3) Acceptance criteria:
    - Project idea and vision are clearly defined
    - All team members share a common understanding of the project goal

Monday, 22. Dezember 2025

Meeting Protocol #2

Sprint Planning (Requirements & Use Cases) Meeting

Attendants / Roles

Luca De Simone – Product owner

Elissar Abdullah – Scrum Master

Delil Celik – Developer

Faycal El Amin – Developer

Discussion Points and Decision:

1) Sprint Goal: The goal of this sprint was to define the system requirements and describe the use cases from the user's perspective.
   - Functional requirements were discussed based on user needs.
   - The final set of three use cases was defined:
     - AR Building Recognition and Historical Information
     - Guided AR Route Navigation
     - App Login with Iris

2) Sprint Backlog:
   - Write textual descriptions for all use cases
   - Define functional and non-functional requirements
   - Assign use cases to team members
   -

3) Acceptance criteria:
   - All use cases are clearly described in text form
   - Requirements reflect the user's expectations
   - Scope of the system is clearly defined

Monday, 05. January 2026

Meeting Protocol #3

Sprint Planning (UML Design)

Attendants / Roles

Elissar Abdullah – Product owner

Faycal El Amin – Scrum Master

Delil Celik – Developer

Luca De Simone – Developer

Discussion Points and Decision:

1) Sprint Goal: The goal of this sprint was to model the defined requirements using UML design.
    - The use case diagram was created based on the previously defined use cases
    - Class, activity, and sequence diagrams were designed to describe the system behaviour
    - It was agreed that UML diagrams must reflect the textual use case descriptions
    - Diagrams were reviewed internally for correctness and consistency
    -
2) Sprint Backlog:
    - Create use case diagram
    - Create class diagram
    - Create activity and sequence diagrams for all use cases
    - Review UML diagrams within the team
    -
3) Acceptance criteria:
    - UML diagrams correctly represent the requirements
    - Diagrams are UML-compliant and consistent
    - All use cases are covered by UML models

Monday, 12. January 2026

Meeting Protocol #4

Sprint Review & Retrospective

Attendants / Roles

Delil Celik – Product owner

Luce De Simone – Scrum Master

Faycal El Amin – Developer

Elissar Abdullah – Developer

Discussion Points and Decision:

1) Sprint Goal: The goal of this sprint was to finalize the project structure and prepare the report for submission
    - The overall structure of the project report was finalized
    - All chapters were aligned with the UML diagrams and requirements
    - Formatting, consistency, and language were reviewed
    - In the retrospective, the team reflected on the process and collaboration
    -
2) Sprint Backlog:
    - Finalize project structure
    - Final proofreading of the report
    - Final consistency check of diagrams and text
    - Prepare final submission
    -
3) Acceptance criteria:
    - Project structure is clear and consistent
    - All requirements content is included

Project is ready for submission

5 Tables

5.1 List of Abbreviations

UC: Use case

POI: Point of interest

FOV: Field of view

GUI: Graphical User Interface

AR: Augmented Reality

5.2 List of Figures

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5.3 AI Prompt Table

Prompt	Result
“A first-person view through AR smart glasses, showing a historic building in an outdoor city square with an AR overlay displaying information about the building”	Figure 6